Research Paradigms:

- Philosophical stance/worldview
- Broad set of beliefs
- Includes ontological and epistemological points of view

Research Paradigms:

- Paradigm: a philosophical/theoretical framework of a scientific discipline that
    - Holds assumptions about social world
    - Tells us how science should be conducted
    - Tells us what constitutes legitimate problems, solutions, criteria of 'proof'

Ontology:

- Greek: "ONTOS" = being/to be
- Branch of philosophy concerned with claims/assumptions about the nature of reality
- What is existence? What is reality?
- Is it stable/ever-changing?
- Are we neutral observers or is reality actively created by us being here?
- What is the nature of existence?

Epistemology:

- Greek: "EPISTIME" = knowledge
- Branch of philosophy concerned with nature and definition of knowledge and truth
- Leads to what one considers as data that is valid and useful
- Gaining knowledge of social reality by examining the relationship between the research and reality
- How can we know about reality?

Methodology:

- The process of finding social reality
- Approach to data collection/analysis
- Finding the right research design
- What procedure do we use to acquire knowledge about social reality

Methods- Methodology- Epistemology- Ontology (ice berg diagram order)

Two Contrasting Paradigms:

- If we believe world/reality are fixed and stable with one true reality, we seek methods that objectively observe and measure systematically
- If we believe reality is actively constructed through perceptions/beliefs, we are more likely, to want to talk to people in order to explore their version of reality

Epistemology: Postivism
- Realism and objectivity
- Valid knowledge, trust generated through scientific processes based on observation/measurement and generalization
- Reality is stable, governed by universal laws, independent of our individual opinions views
- Value-free study through quantitative scientific methods
- Too much theory distracts from the data

Epistemology: Interpretivism
- Subjectivity and constructivism
- Study through qualitative methods. Focus on language, symbols, historical context, culture
- We embrace our subjective influence, don’t aim to remain objective/neutral
- Reality is a combo of multiple, realities that depend on individual people
    - There is no single, unified, stable and objective reality that is waiting to be discovered
    - Our views/attitudes/beliefs are important in constructing multiple realities
- Stresses importance of our perceptions, individual views

Epistemology: Critical Realism (CR)
- Research is not an empirical discipline (not everything can be measured)
- Reality exists independently of individuals’ perceptions
- Critique of positivism: theory and philosophy underlie our questions/methods
- Social world cannot be mathematically represented like the natural world
- Social world is made up of social behaviopirs/isntituisno.cutlure.encomies/lanauges/bleifs/practces (much more complex than can be represented in mathematical formulas)
- Critique of interpretivism: privileges the perceptions of social actors
- CR focuses on that which we can theorise, including things we can't directly measure/see in daily life
- Structuralism

Which study design is appropriate for my research topic/question?
1. Qualitative:
- Little known, topic not fully understood yet
- Need to understand social world/context from perspective of partipaints
- Seeking meaning, understanding
- Seeking in-depth exploration of topic

2. Quantitative:
- Variables influence outcome are understood can be controlled by researcher
- Outcome is easily quantified, can be measured using standardised assessment tools

Quantitative Approaches:
1. Experimental/intervention study: assigns participants to receive a particular exposure, testing effect of intervention/treatment
2. Observational/survey study: does not intentionally expose participants to intervention/ask them to change their behaviours, for generalised from a sample to a population

Epidemiology: study of distribution/determinants of health in human populations
- Descriptive epidemiology studies: observational studies that quantify how often various health-related exposures and outcomes occur in a pop (answer questions like what, who, where, when)
- Analytic epidemiology studies: seek to identify risk/protective factors for various adverse health outcome or to test effectiveness of interventions intended to improve health status (answer questions like why and how)

How study populations are selected:
- Most clinical and population health studies have a research question that emphasizes one particular exposure, disease, or population
- Select participants based on disease status, select participants who represent a population, select participants based on exposure status

Timing of Studies:
- Cross-sectional and case-control studies: all data can be collected at one time.
- Case series, qualitative study:
- Most studies collect all data at one time, but some follow participants forward in time
- Cohort and experimental studies: participants must be followed forward in time

Quantitative Designs:
1. Randomized Controlled Trial: Participants- stratification- randomization- experimental/control group = outcome
2. Cohort Design: Participants- Exposure to/no exposure to intervention = outcome
3. Before-After Design: Group/individual- assessment and baseline evaluation- intervention = outcome
4. Case Control Design (Retrospective): Participants with/without Outcome of Interest- Assessment of previous exposure to intervention or casual factor- compare b/w groups

5. Cross Sectional Design: Participants- measurement of outcomes and other factors at the same time

Qualitative Approaches:
- Ethnography: “What is the culture of a group of people”
- Phenomenology: “What is it like to have a certain experience”
- Grounded theory: theory construction and verification
- Participatory Action Research: social change through research

Qualitative Methods:
- Participant observation
- Interviews
- Focus groups
- Document Review
- Open-ended surveys
- historical/archive review
- Social mapping

Qualitative vs Quantitative Research:
- Review comparison tables on slides
- Qualitative research contributes to in-depth understanding of context in which the phenomena under study takes place
- Quantitative research generates reliable population-based and generalizable data

Key Differences:
- Positivism asks ‘How many?’ or ‘Does X cause Y?’: looks for **universal truths**
- Interpretivticms asks ‘Why?’, or ‘How does this feel?’: looks for context-specific insights
- As health scholars, we need both. Understanding a health issue holistically means we might need biomedical data (positivism) and the social/cultural context (interpretivism)

Combining Qualitative and Quantitative Research:
- Advantages of mixed methods:
    - One approach can be used to inform the other
    - Increased validity (confirmation of resulting by using diff data sources)
    - Complementarity (adding words to numbers, vice versa)
    - Creating new lines of thinking by the emergence of fresh perspectives/contradictions

Which one first?

- Quant to Qual: Variables for survey already well-defined, focus groups are used to help understand factors leading to a particular pattern of quant results
- Qual to Quant: phenomenon of interest is not well documented/understood. Themes emerging from focus groups define variables to be measured in survey

Criteria for assessing validity of quantitative studies:

- Reliability: data collection instruments used provided accurate, consistent measures of attributes/variables researcher intended to measure
- Internal validity: research design adequately controlled for extraneous variables, eliminated plausible rival explanations for research findings
- External validity: researcher employed a sampling strategy that permits the generalisation of the results beyond the specific research participants, research setting and time period

Threats to Validity: A Quantitative Concern

- Sources of **bias** are ways in which the study is conducted that affect the results of a study in one direction (either favours the treatment group or the control group)
- **Confounders** are other factors which may be operating in conjunction with the intervention and offer additional explanations for results of intervention

Criteria for Assessing Qualitative Studies:

- Credibility: results, conclusions make sense from research participants' perspectives
- Transferability: sufficient information provided about context, participants, and assumptions underlying the research so reader can assess potential transferability of findings to other (similar) settings
- Dependability: detail is provided about each step in the research process and contextual factors that influenced decisions about research process
- Confirmability: critical examination of the 'chain of evidence' indicates that the researcher's conclusions are supported by the data

Trustworthiness: A Qualitative Concern

- Researchers should report on methods employed to ensure the quality of the findings
- Triangulation: multiple sources of data, data collection strategies, researcher perspectives, theories
- Member checking: participants verified interpretations of data